

ECON 1550 International Finance

# Price Levels and the Exchange Rate in the Long Run

# Law of One Price (LOOP)

- The LOOP holds if for every good  $i$  with dollar price  $P_{US}^i$  and euro price  $P_E^i$ , we have

$$P_{US}^i = E_{\$/\epsilon} \times P_E^i$$

- This is a theory of exchange rate determination:

$$E_{\$/\epsilon} = \frac{P_{US}^i}{P_E^i}$$

# Purchasing Power Parity (PPP)

- PPP holds if

$$P_{US} = E_{\$/\epsilon} \times P_E$$

where  $P_{US}$  is the US price level and  $P_E$  is the euro area price level

- This is also a theory of exchange rate determination:

$$E_{\$/\epsilon} = \frac{P_{US}}{P_E}$$

# Relative PPP

- Relative PPP holds if

$$\frac{E_{\$/\epsilon,t} - E_{\$/\epsilon,t-1}}{E_{\$/\epsilon,t-1}} = \pi_{US,t} - \pi_{E,t}$$

where  $\pi_t$  is inflation  $\pi_t = P_t/P_{t-1} - 1$

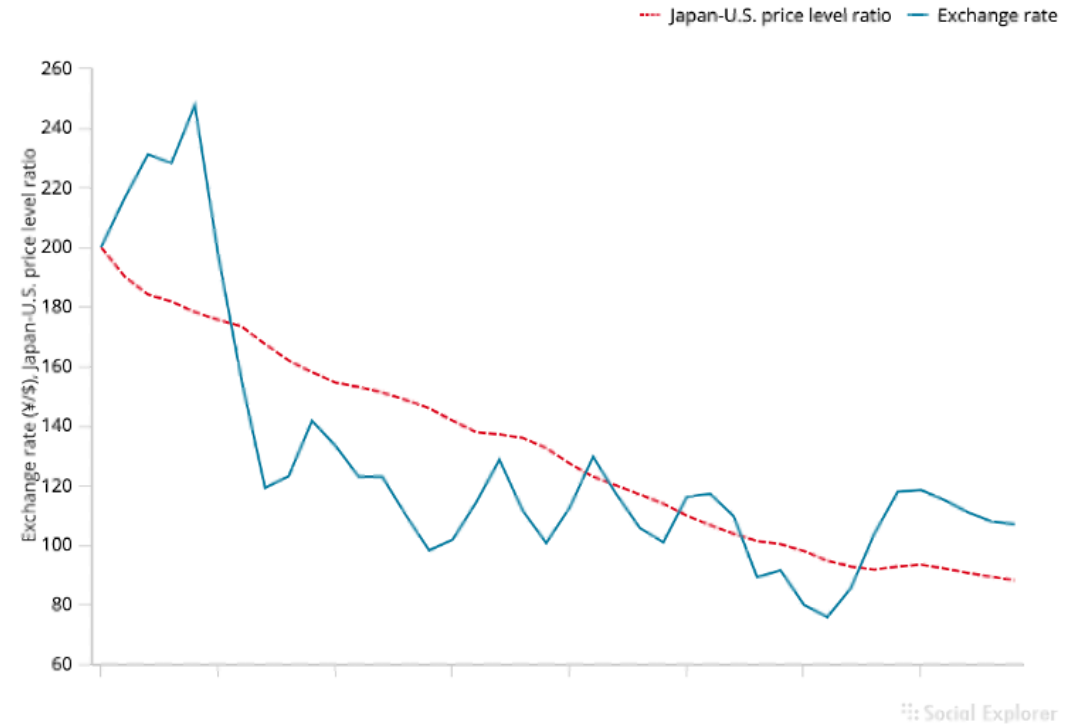
- This is also a theory of exchange rate determination

$$E_{\$/\epsilon,t} = (\pi_{US,t} - \pi_{E,t} + 1)E_{\$/\epsilon,t-1}$$

Relative PPP  
does not hold  
very well in  
the real world

Figure 16-2

The Yen/Dollar Exchange Rate and Relative Japan-U.S. Price Levels, 1980–2019



The graph shows that relative PPP does not track the yen/dollar exchange rate during 1980–2015.

**Source:** IMF, *International Financial Statistics*. Exchange rates and price levels are end-of-year data.

# Problems With PPP

- Price levels of different countries report different baskets of goods
  - E.g. GDP deflator is the basket of goods in GDP for each country
- Deviations from perfectly competitive frictionless markets
  - Transport costs and trade barriers
  - Monopoly power

# Expected Relative PPP

- Expected relative PPP (or relative PPP in expectations)

$$\frac{E_{\$/\epsilon}^e - E_{\$/\epsilon}}{E_{\$/\epsilon}} = \pi_{US}^e - \pi_E^e$$

where  $\pi^e$  is *expected* inflation  $\pi^e = P^e/P - 1$

- This is also a theory of exchange rate determination

$$E_{\$/\epsilon} = \frac{E_{\$/\epsilon}^e}{\pi_{US}^e - \pi_E^e + 1}$$

# Relationships

- LOOP  $\Rightarrow$  PPP  $\Rightarrow$  Relative PPP  $\Rightarrow$  Expected relative PPP
- Expected relative PPP + UIP  $\Rightarrow$  Fisher effect

$$\frac{E_{\$/\epsilon}^e - E_{\$/\epsilon}}{E_{\$/\epsilon}} = \pi_{US}^e - \pi_E^e \quad \text{and} \quad R_{\$} = R_{\epsilon} + \frac{E_{\$/\epsilon}^e - E_{\$/\epsilon}}{E_{\$/\epsilon}}$$

imply

$$R_{\$} - R_{\epsilon} = \pi_{US}^e - \pi_E^e$$

# Common variables across all models

## Exogenous variables

| Variable       | Description               |
|----------------|---------------------------|
| $R^*$          | Foreign interest rate     |
| $M^s, M^{s*}$  | Money supply              |
| $g_M, g_{M^*}$ | Money supply growth rates |

## Endogenous variables

| Variable      | Description         | Equation                   | Type of equation           |
|---------------|---------------------|----------------------------|----------------------------|
| $P$           | Price level         | $M^s/P = L(R, Y)$          | Behavioral + eq. condition |
| $P^*$         | Foreign price level | $M^{s*}/P^* = L(R^*, Y^*)$ | Behavioral + eq. condition |
| $\pi, \pi^*$  | Inflation           | $\pi = g_M - g_L$          | Definition                 |
| $r^e, r^{e*}$ | Real interest rates | $r^e = R - \pi^e$          | Definition                 |

# Model 1: PPP

## Exogenous variables

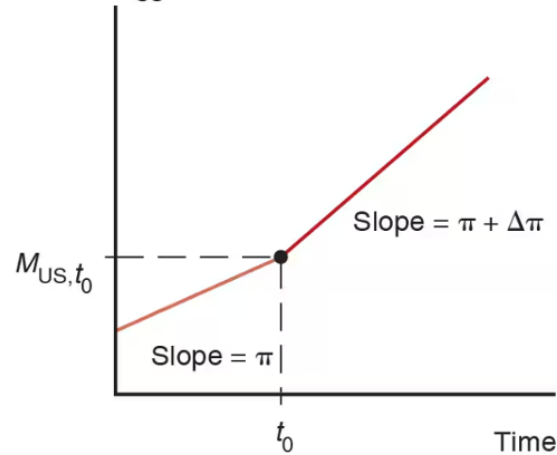
| Variable | Description            |
|----------|------------------------|
| $R$      | Domestic interest rate |
| $Y, Y^*$ | Real incomes           |

## Endogenous variables

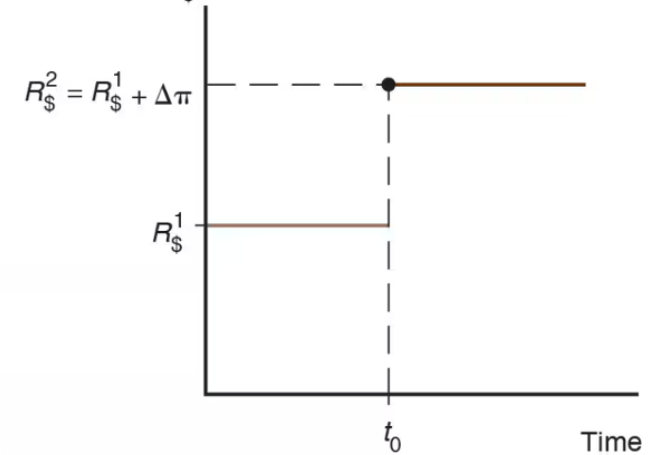
| Variable | Description   | Equation    | Type of equation |
|----------|---------------|-------------|------------------|
| $E$      | Exchange rate | $E = P/P^*$ | Behavioral       |

# Changes in growth rate of money supply

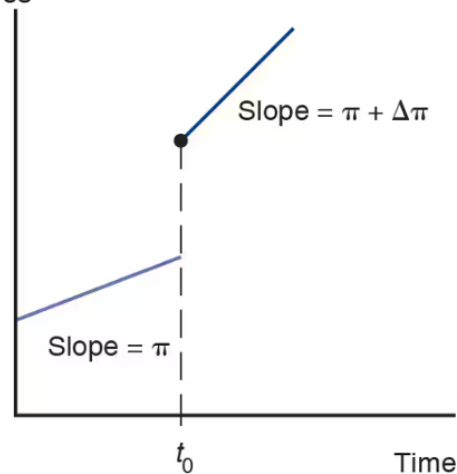
(a) U.S. money supply,  $M_{US}$



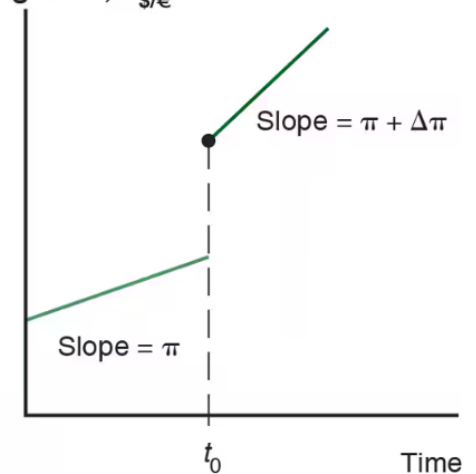
(b) Dollar interest rate,  $R_{\$}$



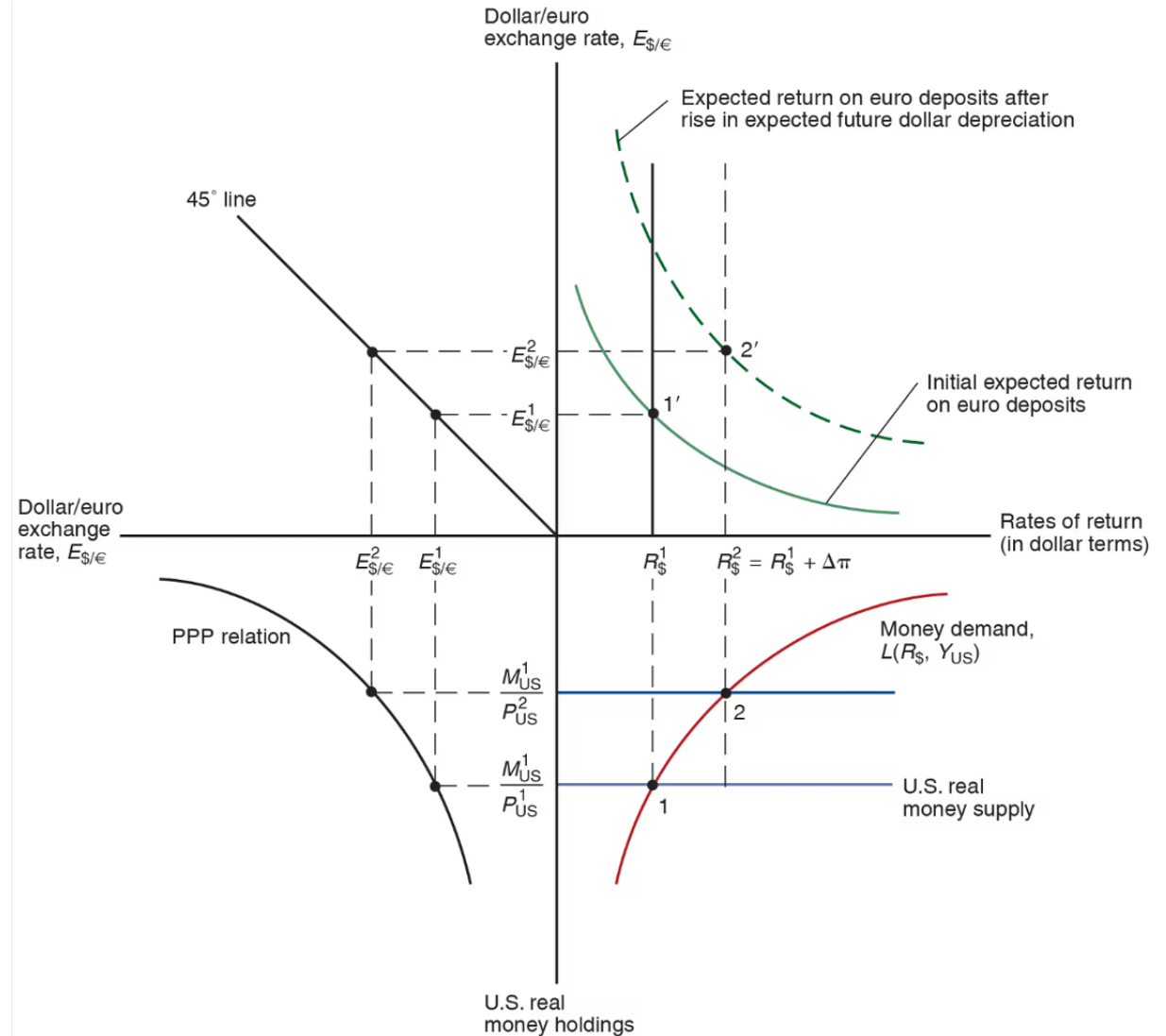
(c) U.S. price level,  $P_{US}$



(d) Dollar/euro exchange rate,  $E_{\$/\epsilon}$



# Changes in growth rate of money supply



# Model 2: PPP<sup>e</sup> + UIP

| Exogenous variables |              | Endogenous variables |                       |                                |                  |
|---------------------|--------------|----------------------|-----------------------|--------------------------------|------------------|
| Variable            | Description  | Variable             | Description           | Equation                       | Type of equation |
| $Y, Y^*$            | Real incomes | $E^e/E - 1$          | Expected depreciation | $E^e/E - 1 = \pi^e - \pi^{e*}$ | Behavioral       |
|                     |              | $R$                  | Interest rate         | $R = R^* + E^e/E - 1$          | Eq. condition    |

# Model 3: real $E$ + relative output

- The real exchange rate :  $q \equiv E \cdot \frac{P^*}{P}$

$$E = 2 \text{ USD/EUR}$$

$$P = 5 \text{ USD/US-GOODS}$$

$$P^* = 3 \text{ EUR/EURO-GOOD}$$

$$q = 2 \frac{\cancel{\text{USD}}}{\cancel{\text{EUR}}} \times 3 \frac{\cancel{\text{EUR}}}{\text{EURO-GOOD}} \frac{1}{\frac{5 \cancel{\text{USD}}}{\text{US-GOOD}}}$$

$$= \frac{6}{5} \frac{\text{US-GOODS}}{\text{EURO-GOODS}}$$

$E \cdot P^*$  = DOLLAR PRICE OF  
FOREIGN GOODS

# Model 3: real $E$ + relative output

- Real exchange rate calculation

$$q = E \cdot \frac{P^*}{P} = \frac{6}{5} \frac{\text{US-GOODS}}{\text{EUR-GOODS}}$$

$$(\text{PPP}) : E = \frac{P}{P^*} \quad \Leftrightarrow \quad q = 1$$

$$\Rightarrow E = q \cdot \frac{P}{P^*} = k \cdot P \quad \text{with } k = \frac{q}{P^*}$$

# Model 3: real $E$ + relative output

$$E = 1 \text{ USD/EUR}$$

↓

$$E = 2 \text{ USD/EUR}$$

↓

EUR CAN BUY

MORE

US GOODS

## Exogenous variables

| Variable | Description            |
|----------|------------------------|
| $RS$     | Relative output supply |
| $q$      | Real exchange rate     |

## Endogenous variables

| Variable | Description           | Equation     | Type of equation |
|----------|-----------------------|--------------|------------------|
| $Y/Y^*$  | Relative output       | $RD(q) = RS$ | Eq. cond.        |
| $E$      | Nominal exchange rate | $E = qP/P^*$ | Definition       |

$$RD\left(\frac{EP^*}{P}\right) = RS$$

(+)

$E \uparrow$  DEPRECIATION OF HOME CURRENCY

$Y \uparrow$  AND  $Y^* \downarrow$

# Model 3 equilibrium

## Determination of the Long-Run Real Exchange Rate

$$\left. \begin{aligned} q &= q^1 \\ q &= \frac{E P^*}{P} \end{aligned} \right\} \text{DETERMINE } E$$

$P, P^*$  DETERMINED  
IN MONEY MKT

$$E = q^1 (P/P^*)$$

